Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech
Code: 105601

Subject: Compiler Design

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct option for the following questions (Answer any seven):* **[2 x 7 = 14]**

(a) Which phase of the compiler is responsible for removing comments and white spaces?
- [✓ (i) Syntax Analysis]
- (ii) Lexical Analysis
- (iii) Semantic Analysis
- (iv) Intermediate Code Generation

(b) A lexical analyzer uses which of the following automata?
- (i) PDA
- (ii) Turing Machine
- [✓ (iii) DFA/NFA]
- (iv) LBA

(c) LL(1) parsers use which parsing technique?
- (i) Bottom-up
- (ii) Shift-reduce
- [✓ (iii) Top-down]
- (iv) Operator precedence

(d) Conflicts in LR parsing include
- (i) Shift-shift
- (ii) Shift-reduce
- (iii) Reduce-reduce
- [✓ (iv) Both b and c]

(e) Type checking ensures
- (i) Syntax correctness
- (ii) Register allocation
- [✓ (iii) Semantic consistency]
- (iv) Pipeline optimization

(f) Pipeline optimization helps reduce
- (i) Syntax errors
- [✓ (ii) Register conflicts]
- (iii) Execution stalls
- (iv) Semantic errors

(g) Polymorphism is implemented using
- (i) NFA
- (ii) Heap
- [✓ (iii) Virtual functions]
- (iv) Stack frames

(h) Which parser construction method is used by YACC?
- (i) LL
- (ii) LALR
- [✓ (iii) General LR]
- (iv) SLR

(i) DAG stands for
- (i) Directed Applicable Graph
- [✓ (ii) Directed Acyclic Graph]
- (iii) Data Analysis Graph
- (iv) Data Allocation Graph

(j) LEX tool is used for
- (i) Parser generation
- [✓ (ii) Lexical analysis]
- (iii) Syntax analysis
- (iv) Intermediate code generation

Q.2 (a) What are the various phases of a compiler? Explain each phase in detail by processing the input "a=(b+c)*(b+c)*2". **[7]**

(b) What is meant by lexical analysis, token, pattern and lexeme? Identify the lexemes that makeup the token in the following program segment. Indicate the corresponding token and pattern. **[7]**

```
Void swap(int i, int j)
int t; t = i ; i = j ; j = t ;
```

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Q.3 (a) Construct set of LR(1) items for given grammar and check whether it is SLR, CLR or LALR. [7]

$S \rightarrow CC$, $C \rightarrow aC$, $C \rightarrow d$

(b) Consider the grammar [7]

$E \rightarrow T E_1$, $E_1 \rightarrow +T E_1 \mid \varepsilon$

$T \rightarrow F T_1$, $T_1 \rightarrow * F T_1 \mid \varepsilon$

$F \rightarrow (E) \mid id$

Find the FIRST and FOLLOW set of all the variables.

Q.4 (a) Consider the following grammar along with translation rules. [7]

$S \rightarrow S_1 \# T$ {S.val=S₁.val*T.val}
$S \rightarrow T$ {S.val=T.val}
$T \rightarrow T_1 \% R$ {T.val=T₁.val÷R.val}
$T \rightarrow R$ {T.val=R.val}
$R \rightarrow id$ {R.val=id.val}

Here # and % are operators and *id* is a token that represents an integer and *id.val* represents the corresponding integer value. Using this translation scheme, compute the value of *S.val* for root of the parse tree for the expression 20#10%5#8%2%2.

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(b) Consider the below grammar, which defines simple arithmetic expressions. [7]

$E \rightarrow E + T \mid E - T \mid T$

$T \rightarrow T * F \mid T / F \mid F$

$F \rightarrow (E) \mid id$

Now modify the above grammar with unconventional rules of precedence and associativity and then rewrite such that,

(a) the expression 3*2+1 will be interpreted as 3* (2+1).

(b) the value of 1-2-3 will be 2 rather than -4.

Q.5 (a) Convert the regular expression (a|b)*ab into NFA, then convert NFA into minimal DFA. [7]

(b) Differentiate between synthesized and inherited attributes with example. [7]

Q.6 (a) What are shift-reduce and reduce-reduce conflicts? Explain with examples. [7]

(b) Differentiate between Top-down and Bottom-up parsing with example. [7]

Q.7 (a) Consider the following C code segment: [7]

```
a = b + c;
e = a + 1;
d = b + c;
f = d + 1;
g = e + f;
```

In a compiler, this code segment is represented internally as a directed acyclic graph (DAG). Draw the DAG representation and find the number of nodes in the DAG.

(b) Explain architecture dependent code optimization techniques with examples. [7]

Q.8 (a) How are object-oriented features compiled in C++/Java-like languages? [7]

(b) Consider the following pseudo-code. [7]

L1: t1=−1	L6: t6=t5*M	L11: t3=t3+1
L2: t2=0	L7: t7=t4+t6	L12: if t3 < M goto L4
L3: t3=0	L8: t8=a[t7]	L13: t2=t2+1
L4: t4=4*t3	L9: if t8 <= max goto L11	L14: if t2 < N gotoL3
L5: t5=4*t2	L10: t1=t8	L15: max=t1

Find the number of basic blocks and the number of instructions in the largest basic block.

Q.9 Write Short notes on ***any two*** of the following: [7 x 2 = 14]

(a) Symbol Table
(b) Peephole Optimization
(c) Activation Record
(d) Three Address Code

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